PRATIK PATIL

DATA SCIENTIST & SYSTEM ENGINEER

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PROFESSIONAL PROFILE

Data Science postgraduate with a Bachelor's in Computer Science and a strong foundation in System Programming. UGC NET qualified (89.28 percentile) with expertise in quantitative finance, RAG architectures, and AI Agents. Highly skilled in leveraging Python, Rust, and Go to build high-performance, scalable machine learning systems and data pipelines.

WORK EXPERIENCE

Freelance Data Scientist & Quant Researcher *Nov 2022 – Present*
Self-Employed *Remote*

- Engineered predictive machine learning models for financial markets, leveraging time-series forecasting to identify high-probability trading signals.
- Designed and backtested algorithmic trading strategies, optimizing for risk-adjusted returns and portfolio allocation.
- Architected scalable data pipelines using Python, SQL, and Rust to process large-scale datasets, generating actionable insights for clients.

Java Developer Intern *Sep 2019 – Nov 2019*
Avalanche Infotech *Indore, India*

- Automated unit testing utilizing JUnit, significantly reducing manual QA time and accelerating release cycles.
- Integrated CI/CD pipelines to streamline deployment and ensure code reliability in Agile environments.
- Collaborated on deploying secure, efficient multi-threaded server-side solutions.

KEY PROJECTS

Veritas: High-Performance Deduplication Engine **Rust**

- Developed an intelligent record deduplication engine to clean massive datasets with sub-second latency.
- Leveraged Rust for memory-safe, high-speed string matching (MinHash/LSH), exposed via a Python REST API, reducing data redundancy by 90%.

Polymorph: Image Processing & ASCII API **Python, ML, HTML/CSS, Go**

- Built a blazing-fast REST API for image conversion and ASCII art generation, integrating Rust modules for compute-intensive tasks.
- Architected the backend in Go to handle high-concurrency requests, with a lightweight HTML/CSS frontend.
- Deployed a containerized architecture with automated testing, ensuring 99.9% uptime.

Log-Query: High-Performance Log Search Tool **Rust**

- Developed a CLI application in Rust to parse and query massive log files for specific keywords with minimal memory overhead.
- Implemented efficient string matching and file I/O to deliver rapid search results across gigabytes of text data.

Analysis Before Wicket: Cricket Analytics Platform **R, Python, HTML/CSS, Go**

- Built a comprehensive data-driven cricket analytics platform to evaluate player performance and match forecasting.
- Engineered the backend in Go for rapid data retrieval, while using R and Python for advanced statistical modeling and forecasting.
- Designed an interactive frontend using HTML/CSS to visualize complex cricket statistics and trends.

EDUCATION

Master of Technology (Data Science) *2020 – 2022*
Devi Ahilya Vishwavidyalaya (DAVV)

Bachelor of Engineering (Computer Science) *2016 – 2020*
Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV)

TECHNICAL SKILLS & INTERESTS

- **Languages:** Python, Rust, Go, SQL, Java, HTML/CSS
- **Databases:** SQLite, MongoDB, MySQL
- **Machine Learning & AI:** Deep Learning, NLP, RAGs, AI Agents, Forecasting
- **Tools & Visualization:** Tableau, Docker, Git, PowerBI
- **Core Interests:** Quantitative Finance, Scientific Computing, System Programming

ACHIEVEMENTS & CERTIFICATIONS

- **UGC NET 2025:** Qualified with 89.28 percentile.
- **UGC NET 2024:** Qualified with 81 percentile.
- **Certifications:** SQL Advanced (HackerRank), Python Intermediate (HackerRank).